Final Metrics

PDF edition of preserved repository document | docs/final_metrics.md

Historical statements and test counts below retain their original scope. Consult the master report for current status.

Actual completion metrics

Machine-readable full metrics: results/coding/emergency_analysis_v1/summary.json.

Mindscape bounded MPS completion evidence

Learning curve: 1920/1,920 episodes, 0 not run. Lockbox: 400/400 condition-task episodes (100 independent tasks). Preserved CPU work remains intact.

All metrics below derive from saved episodes. Missing cells are not zero scores. CPU float32 and MPS reduced-precision strata are kept separate. Supervision and interaction budgets differ across conditions; architecture-only causality is not established.

Condition	Split	Completed/planned	Success	95% Wilson interval
model_only	test	50/50	16.0%	[0.0833742067803340 4, 0.285142160630349 9]
model_only	ood_test	50/50	24.0%	[0.1429739139699173, 0.3741268375794292]
structured	test	50/50	100.0%	[0.9286524008666414, 1.0]
structured	ood_test	50/50	44.0%	[0.3116219921125717, 0.5769397197834315]
mindscape_b	test	50/50	90.0%	[0.7863976856252035, 0.9565242350681096]
mindscape_b	ood_test	50/50	48.0%	[0.3479713528657804 6, 0.614882551099553 9]
mindscape_c	test	50/50	98.0%	[0.8950455641036219, 0.9964607407283539]
mindscape_c	ood_test	50/50	36.0%	[0.2413874965184674 1, 0.498589831238873 07]

Observed complete-cell sample thresholds and all partial cells are in results/coding/emergency_analysis_v1/summary.json. DER is undefined for a causal architecture comparison.

No zero-budget control was run.

Task-paired confidence intervals are descriptive and unadjusted for multiple comparisons. Transition validity is conditional on recorded actions; malformed model proposals are reported separately.

Observed self-correction from saved offline first-edit assessments: [{"condition": "mindscape_b", "split": "test", "completed_episodes": 50, "first_edit_failures": 0, "later_terminal_successes": 0, "conditional_self_correction_rate": null, "wilson_ci95": null, "definition": "Terminal success after an incorrect first accepted edit; first-edit assessment occurs offline after policy termination, never as hidden model feedback", "caution": "Descriptive within-episode self-correction; no injected-fault recovery intervention or causal memory/planner attribution"}, {"condition": "mindscape_b", "split": "ood_test", "completed_episodes": 50, "first_edit_failures": 38, "later_terminal_successes": 12, "conditional_self_correction_rate": 0.3157894736842105, "wilson_ci95": [0.1908460367559618, 0.4745575989060954], "definition": "Terminal success after an incorrect first accepted edit; first-edit assessment occurs offline after policy termination, never as hidden model feedback", "caution": "Descriptive within-episode self-correction; no injected-fault recovery intervention or causal memory/planner attribution"}, {"condition": "mindscape_c", "split": "test", "completed_episodes": 50, "first_edit_failures": 1, "later_terminal_successes": 0, "conditional_self_correction_rate": 0.0, "wilson_ci95": [0.0, 0.7934506856227626], "definition": "Terminal success after an incorrect first accepted edit; first-edit assessment occurs offline after policy termination, never as hidden model feedback", "caution": "Descriptive within-episode self-correction; no injected-fault recovery intervention or causal memory/planner attribution"}, {"condition": "mindscape_c", "split": "ood_test", "completed_episodes": 50, "first_edit_failures": 32, "later_terminal_successes": 0, "conditional_self_correction_rate": 0.0, "wilson_ci95": [0.0, 0.1071791982550706], "definition": "Terminal success after an incorrect first accepted edit; first-edit assessment occurs offline after policy termination, never as hidden model feedback", "caution": "Descriptive within-episode self-correction; no injected-fault recovery intervention or causal memory/planner attribution"}]. This is separate from unrun injected-fault recovery tests.

Operational answer metrics: {"definition": "Successful terminal program AND independent full-trajectory replay agreement; no semantic hallucination detector", "denominator": 400, "grounded_successful_answers": 228, "terminal_incorrect_answers": 172, "semantic_hallucination_rate": null, "semantic_hallucination_status": "Not measured by this program-repair protocol", "unverified_episodes": 0, "grounded_successful_answer_rate": 0.57, "unsupported_by_terminal_tests_rate": 0.43}. Failed terminal tests are unsupported program answers under this oracle; they are not a measured semantic hallucination rate.

Original frozen configuration SHA-256: 11d2a6b5bef6517a5cae10dca46f1e94bfe2db3504afd3942434fcb78f496bb0. Seeds: 11/23/37 for learning; seed 11 and budget 100 for lockbox. Exact checkpoint/source hashes are in the frozen protocol; host/runtime and training compute are in the machine-readable summary.

Actual acceleration measurements


```
{
  "cpu_seconds_per_episode": 111.3537078230629,
  "cpu_reference_scope": "Preserved fresh CPU float32 episodes; not rerun",
  "mps_throughput_seconds_per_episode": 27.54043339325017,
  "mps_mean_episode_wall_seconds": 372.8890333698155,
  "throughput_speedup": 4.043280882077708,
  "mps_episodes_per_hour": 130.7168971742586,
  "selected_batch_size": 16,
  "precision": "float16",
  "load_seconds": 22.024639666007715,
  "peak_process_rss_bytes": 3421274112,
  "mps_allocated_bytes": 3333364480,
  "mps_driver_bytes": 12526354432,
  "profiling_seconds": 600.09845495224,
  "interpretation": "Historical CPU vs fresh batched MPS throughput; per-case latency
reported separately. Arithmetic precision differs; learning-curve hardware strata remain
disclosed."
}
```


One phase-boundary memory release: {"before_driver_bytes": 10687807488, "after_driver_bytes": 4346019840, "live_allocated_bytes": 3112987904, "reason": "Single profiling/evaluation boundary; no per-operation cache clearing", "configuration_reused": true, "time_unix": 1791468859.762655}.

CPU reference episodes were previously measured with uncached inference; they were not rerun. MPS sec/episode is amortized throughput, distinct from per-episode latency while waiting for batches. Device tensors and model placement are checked; unsupported MPS operations fail with fallback disabled. Native allocation reported by the user is verified, but inference use is only established by native run outputs.

Replay and latency


```
{
  "replay": {
    "episodes": 400,
    "planned": 400,
    "final_outcome_agreement": 1.0,
    "structured_evidence_agreement": 1.0,
    "state_agreement": 1.0,
    "state_comparisons": 2584,
    "tool_result_agreement": 1.0,
    "tool_result_comparisons": 1092
  },
  "latency": [
    {
      "condition": "model_only",
      "episodes": 4,
      "planned": 4,
      "measurement_source": "Dedicated cache-disabled stage benchmark",
      "device_precision_counts": {
        "mps/float16": 4
      },
      "actual_batch_sizes": [
        1
      ],
      "peak_process_rss_bytes": 3802660864,
      "peak_mps_live_bytes": 3132599552,
      "peak_mps_driver_bytes": 4389109760,
      "peak_cuda_allocated_bytes": null,
      "peak_cuda_reserved_bytes": null,
      "sequential_episodes_per_hour": 1475.2094699714482,
      "percentiles": {
        "total": {
          "p50": 2.5287001245001193,
          "p95": 2.9951699749499765,
          "n": 4
        },
        "ttft": {
          "p50": 0.10764597949992094,
          "p95": 0.8070860983006238,
          "n": 4
        },
        "response": {
          "p50": 2.2023833544999434,
          "p95": 2.6520218813000156,
          "n": 4
        },
        "model_generation": {
          "p50": 2.2023833544999434,
          "p95": 2.6520218813000156,
          "n": 4
        },
        "environment_execution": {
          "p50": 0.00013612449993161135,
          "p95": 0.0004516401001183111,
          "n": 4
        },
        "verification": {
          "p50": 0.3319560420000016,
          "p95": 0.3436878607001745,
          "n": 4
        },
```

```
        "test_execution": {
          "p50": 0.3318179789998794,
          "p95": 0.34356011874992876,
          "n": 4
        }
      },
      "caution": "Descriptive sample; stage timers overlap. TTFT is model first-token time
 after CPU input encoding. Response is per-call wall time including encoding/decoding in
dedicated measurements; total is multi-step episode wall time. Batched smoke model
durations are shared across requests. Unmeasured timers are not inferred."
    },
    {
      "condition": "structured",
      "episodes": 4,
      "planned": 4,
      "measurement_source": "Dedicated cache-disabled stage benchmark",
      "device_precision_counts": {
        "mps/float16": 4
      },
      "actual_batch_sizes": [
        1
      ],
      "peak_process_rss_bytes": 3802660864,
      "peak_mps_live_bytes": 3134778624,
      "peak_mps_driver_bytes": 4431052800,
      "peak_cuda_allocated_bytes": null,
      "peak_cuda_reserved_bytes": null,
      "sequential_episodes_per_hour": 1252.859663165493,
      "percentiles": {
        "total": {
          "p50": 2.891284770499624,
          "p95": 3.226074864299926,
          "n": 4
        },
        "ttft": {
          "p50": 0.3468195834998369,
          "p95": 0.3796293562498249,
          "n": 4
        },
        "response": {
          "p50": 2.5587374795004507,
          "p95": 2.895561100049872,
          "n": 4
        },
        "model_generation": {
          "p50": 2.5587374795004507,
          "p95": 2.895561100049872,
          "n": 4
        },
        "environment_execution": {
          "p50": 0.0006867300003250421,
          "p95": 0.0008416026992108527,
          "n": 4
        },
        "verification": {
          "p50": 0.3316854585004876,
          "p95": 0.33450531509984105,
          "n": 4
        },
        "test_execution": {
          "p50": 0.33154877100014346,
          "p95": 0.3343919018002907,
          "n": 4
        }
      },
      "caution": "Descriptive sample; stage timers overlap. TTFT is model first-token time
 after CPU input encoding. Response is per-call wall time including encoding/decoding in
dedicated measurements; total is multi-step episode wall time. Batched smoke model
durations are shared across requests. Unmeasured timers are not inferred."
    },
    {
      "condition": "mindscape_b",
```

```
      "episodes": 4,
      "planned": 4,
      "measurement_source": "Dedicated cache-disabled stage benchmark",
      "device_precision_counts": {
        "mps/float16": 32
      },
      "actual_batch_sizes": [
        1
      ],
      "peak_process_rss_bytes": 3802660864,
      "peak_mps_live_bytes": 3136957696,
      "peak_mps_driver_bytes": 4506550272,
      "peak_cuda_allocated_bytes": null,
      "peak_cuda_reserved_bytes": null,
      "sequential_episodes_per_hour": 95.38661046707223,
      "percentiles": {
        "total": {
          "p50": 35.924512353999944,
          "p95": 44.30880428300006,
          "n": 4
        },
        "ttft": {
          "p50": 1.1655523545000506,
          "p95": 1.6721938956499798,
          "n": 32
        },
        "response": {
          "p50": 4.590948958500576,
          "p95": 5.841075649850472,
          "n": 32
        },
        "model_generation": {
          "p50": 35.228675708500305,
          "p95": 43.44583151340143,
          "n": 4
        },
        "environment_execution": {
          "p50": 0.0012617284996849776,
          "p95": 0.0015499980499498632,
          "n": 4
        },
        "verification": {
          "p50": 0.3748883539997223,
          "p95": 0.47029650980048243,
          "n": 4
        },
        "test_execution": {
          "p50": 0.7139516665001793,
          "p95": 0.8574064588998226,
          "n": 4
        }
      },
      "caution": "Descriptive sample; stage timers overlap. TTFT is model first-token time
 after CPU input encoding. Response is per-call wall time including encoding/decoding in
dedicated measurements; total is multi-step episode wall time. Batched smoke model
durations are shared across requests. Unmeasured timers are not inferred."
    },
    {
      "condition": "mindscape_c",
      "episodes": 4,
      "planned": 4,
      "measurement_source": "Dedicated cache-disabled stage benchmark",
      "device_precision_counts": {
        "mps/float16": 32
      },
      "actual_batch_sizes": [
        1
      ],
      "peak_process_rss_bytes": 3802660864,
      "peak_mps_live_bytes": 3139136768,
      "peak_mps_driver_bytes": 4514938880,
      "peak_cuda_allocated_bytes": null,
```

```
      "peak_cuda_reserved_bytes": null,
      "sequential_episodes_per_hour": 117.28219079755843,
      "percentiles": {
        "total": {
          "p50": 31.74315270850002,
          "p95": 32.581375693700465,
          "n": 4
        },
        "ttft": {
          "p50": 1.1719996460001312,
          "p95": 1.6940761644004851,
          "n": 32
        },
        "response": {
          "p50": 3.9009514585000034,
          "p95": 4.31798346860005,
          "n": 32
        },
        "model_generation": {
          "p50": 31.05320908349904,
          "p95": 31.83074219474984,
          "n": 4
        },
        "environment_execution": {
          "p50": 0.0014917294993210817,
          "p95": 0.0035893313990072776,
          "n": 4
        },
        "verification": {
          "p50": 0.3700127080001039,
          "p95": 0.38461931249967163,
          "n": 4
        },
        "test_execution": {
          "p50": 0.6868583330001456,
          "p95": 0.7454004454003552,
          "n": 4
        }
      },
      "caution": "Descriptive sample; stage timers overlap. TTFT is model first-token time
 after CPU input encoding. Response is per-call wall time including encoding/decoding in
dedicated measurements; total is multi-step episode wall time. Batched smoke model
durations are shared across requests. Unmeasured timers are not inferred."
    }
  ]
}
```


Preserved public benchmark

0.5B: 94/164 (57.32%); 1.5B: 98/164 (59.76%). Greedy one-sample full-module generation, 512-token cap, CPython 3.14.7 WASI. Foundation pretraining exposure is unknown. These local results do not exceed historical GPT-4 67.0% or Gemini Ultra 74.4%, and protocols differ. See docs/historical_references.md for pinned primary sources.

Dedicated latency: 16/16 fresh sequential cases; actual batch sizes and memory are reported above. Sequential response latency is distinct from frozen batch-16 evaluation throughput.

Complete suite: {'tests': 136, 'failures': 0, 'errors': 0, 'skipped': 0, 'pytest_passed_tests': 130, 'subtests_passed': 6, 'seconds': 22.683}. Local fairness/leakage assertions: 10158, PASS. Contamination-free status remains unknown. All 56 retained adapters validated. SQL environment implemented; SQL model study deferred. Recovery and memory/dream ablations not run; their claims remain inconclusive.